

System Handbook

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1. Purpose

This handbook is the primary technical and operational baseline for the UAIL IT Dashboard. It replaces the earlier split across requirements, architecture, security, and deployment notes with one maintained source that matches the current implementation.

2. Product Summary

The UAIL IT Dashboard is an internal LAN-hosted operational wallboard with:

- an operator surface on 21060
- an admin surface on 21061
- a shared Next.js app
- a loopback API gateway
- collectors for Nutanix, SolarWinds, and Symphony HSD
- SQLite as the active runtime store
- optional PostgreSQL for mirror and control-plane use

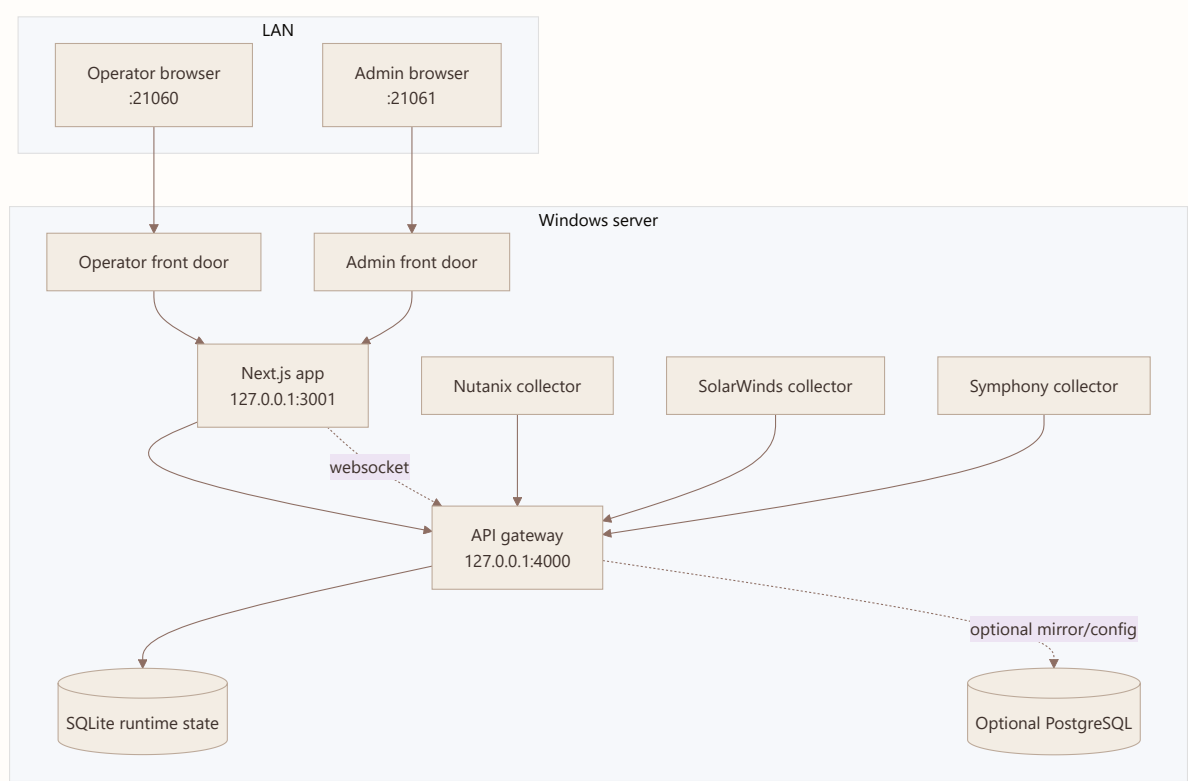
The system is designed around one rule: show only real values from a known source, and keep the last confirmed values visible with their last sync time when a source fails.

3. Users And Surfaces

Surface	Port	Primary Users	Purpose
Operator	21060	Operations wallboard viewers	Live monitoring, filtering, drill-free situational awareness
Admin	21061	Tech-Unit IT admins	Service control, source configuration, session recovery, documentation access

Both surfaces use the same web application and differ by front-door routing, login flow, and authorization.

4. Runtime Topology



5. Functional Scope

5.1 Operator Dashboard

The operator dashboard presents:

- HCI cluster metrics
- HSD backlog and SLA metrics
- network link status and utilization
- server fleet status by platform and OS

Design priorities:

- one-page wallboard for 1920x1200
- usable secondary layout for 1920x1080
- responsive mobile and portrait fallback without overlapping cards
- concise data-link status per card or section

5.2 Admin Console

The admin console provides:

- PM2-backed service state and restart controls
- live HSD and SolarWinds session validation
- server-local HSD reauthentication and legacy-profile import where applicable
- source endpoint and credential settings
- embedded help/documentation PDFs

6. Data Sources And Source Of Truth

Domain	Primary Source	Notes
HCI cluster	Nutanix	Direct Prism API polling
HCI-backed servers	Nutanix	Primary source whenever Nutanix is healthy
On-prem servers	SolarWinds 45	Used for non-HCI server telemetry
Network links	SolarWinds 46	Used for carrier and SDWAN telemetry
HSD backlog and SLA	Symphony HSD	Browser-authenticated collector

6.1 Overlap Rule

- Nutanix is authoritative for any server present in both Nutanix and SolarWinds.
- SolarWinds becomes fallback for Nutanix-backed servers only after Nutanix has been stale for more than 10 minutes and SolarWinds telemetry exists.
- During fallback, the UI must indicate the fallback source rather than presenting the value as if it were still Nutanix-derived.

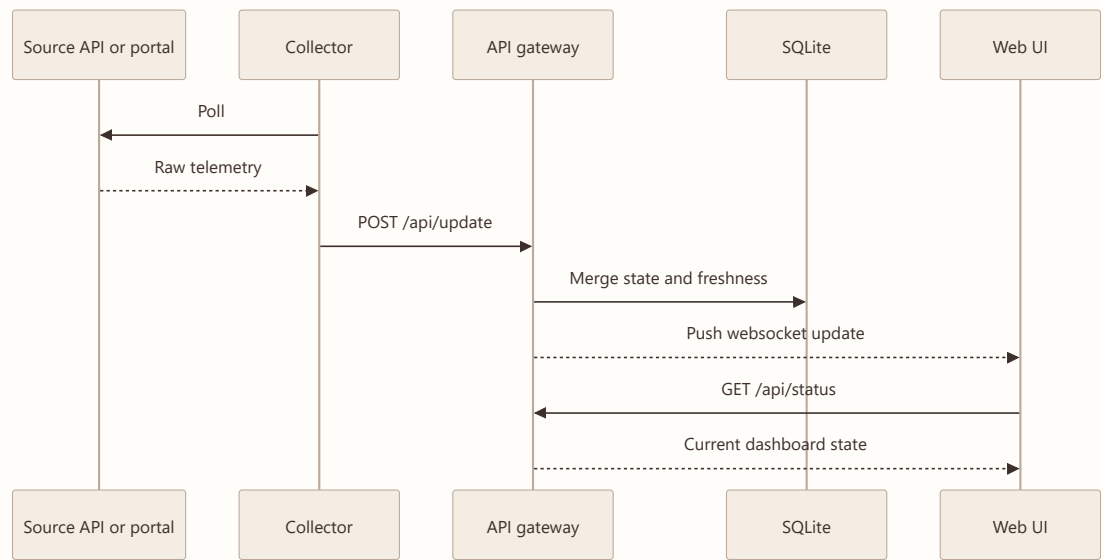
6.2 Failure Rule

- The dashboard must not fabricate normal operating values.
- On collection failure, keep the last successful values visible.
- Show the last synced time and collector/link state for the affected section.

7. Service Inventory

Service	Role	Exposure
dashboard-frontdoor-operator	Operator reverse proxy	LAN
dashboard-frontdoor-admin	Admin reverse proxy	LAN
dashboard-ui	Shared Next.js app	Loopback
api-gateway	Status API, admin APIs, WebSocket feed	Loopback
nutanix-collector	Nutanix polling	Host-local
solarwinds-collector	SolarWinds 45/46 Playwright collection	Host-local
symphony-collector	Symphony HSD Playwright collection	Host-local

8. Collection And Storage Model



Current storage posture:

- SQLite is the active runtime state store.
- PostgreSQL is optional and used only when explicitly configured for mirrored state or stronger control-plane storage.
- Collector target settings and secrets can be stored in the encrypted local runtime store when PostgreSQL is not enabled.

9. Reliability Model

9.1 Freshness Metadata

Each major section tracks:

- poll interval
- last attempt
- last success
- last error

9.2 Auto-Heal Model

Deployed recovery is based on:

- PM2 restart behavior for crashed services
- `pm2 save`
- startup restore via scheduled task and `pm2 resurrect`
- admin service controls for targeted restarts

9.3 Session-Dependent Sources

SolarWinds and HSD still depend on browser-authenticated sessions. Admin users must treat expired sessions as a source failure, not as a UI-only issue.

10. Deployment Baseline

10.1 Supported Host

- Windows Server 2019 or later
- dedicated VM or dedicated application server
- Edge installed
- LAN reachability to Nutanix, SolarWinds, and HSD

10.2 Packaging Paths

- Installer: `deployment/installer/output/utkal-it-dashboard-setup.exe`
- Offline bundle: `deployment/release/utkal-it-dashboard-offline-server-bundle-2026-07-19.zip`
- Staged fallback payload: `deployment/staging/current`

10.3 Core Runtime Paths

- install/runtime root: `C:\ProgramData\UAIL\ITDashboard`
- HSD storage state:
`C:\ProgramData\UAIL\ITDashboard\sessions\symphony\symphony-storage-state.json`
- HSD interactive Edge profile:
`C:\ProgramData\UAIL\ITDashboard\sessions\symphony\interactive-edge-profile`
- local collector settings: `C:\ProgramData\UAIL\ITDashboard\config\collector-settings.json`

10.4 Deployment Validation

Run the packaged smoke test after staging or installer rebuilds:

```
powershell -NoProfile -ExecutionPolicy Bypass -File  
deployment\tools\validate-offline-deployment.ps1
```

This provisions an isolated temporary install, starts the stack on `22160` and `22161`, validates the operator and admin login flows, checks the admin services API, and tears the test install back down.

11. Security Posture

11.1 Current Controls

- separate operator and admin front doors
- separate signed cookies for operator and admin sessions
- admin-only APIs protected in the Next.js layer
- loopback restriction for internal services and runtime config endpoints
- encrypted local secret store supported via `SECRET_STORE_PASSPHRASE`

11.2 Known Gaps

- local password-only login is simpler than enterprise SSO
- `/api/update` still relies on host and network isolation rather than a dedicated application-layer collector key
- front-door traffic is HTTP unless network controls or TLS termination are added outside the app
- Nutanix TLS verification is disabled because of upstream certificate conditions
- SolarWinds and HSD session material remains on disk by design

11.3 Required Compensating Controls

- deploy on a dedicated internal Windows server or VM
- expose only 21060 and 21061 on approved internal segments
- keep 3001 and 4000 loopback-only
- ACL-protect C:\ProgramData\UAIL\ITDashboard
- use dedicated upstream service credentials
- back up runtime configuration and session-state files securely

12. Acceptance Baseline

The current implementation is considered aligned when:

- operators can log into 21060 and view all enabled sections
- admins can log into 21061 and manage services, sessions, and source settings
- section link state reflects real collector freshness and last-sync data
- Nutanix-backed servers revert to Nutanix when Nutanix data becomes healthy again after fallback
- network telemetry is sourced from SolarWinds 46
- HSD bucket counts and special queues reflect live portal mappings

13. Current Constraints

- Nutanix uses non-ideal internal TLS certificates
- SolarWinds and HSD are still browser-session dependent
- SQLite is intentionally the active store for deployment simplicity
- the local login model is acceptable only for a controlled internal deployment

14. Maintained References

- [Documentation Index](#)
- [Operations Guide](#)
- [Project Timeline](#)
- [Executive Summary Pack](#)
- [Deployment Workspace Notes](#)